Experiment 6: Total Purchase Cost with Discount and Tax

Aim

To calculate the total cost of a customer's purchase by applying discounts and taxes using arithmetic operations in Python.

Algorithm

1. **Start the program.**
2. **Define lists containing item prices and quantities.**
3. **Calculate the subtotal by multiplying each item's price by its quantity and adding the results.**
4. **Calculate the discount amount using the discount rate.**
5. **Subtract the discount from the subtotal.**
6. **Calculate the tax on the discounted amount.**
7. **Add the tax to the discounted subtotal.**
8. **Display the subtotal, discount, tax, and final cost.**
9. **Stop the program.**

Code

```
prices = [100, 200, 150]

quantities = [2, 1, 3]

discount_rate = 10  # 10%

tax_rate = 5       # 5%

subtotal = sum(
   price * quantity
   for price, quantity in zip(prices, quantities)
)

discount = subtotal * discount_rate / 100

discounted_total = subtotal - discount

tax = discounted_total * tax_rate / 100

total_cost = discounted_total + tax
```

print("Subtotal:", f"{subtotal:.2f}")

print("Discount:", f"{discount:.2f}")

print("Amount after discount:", f"{discounted_total:.2f}")

print("Tax:", f"{tax:.2f}")

print("Final total cost:", f"{total_cost:.2f}")

Input

Item prices = [100, 200, 150]

Quantities = [2, 1, 3]

Discount rate = 10%

Tax rate = 5%

Output

Subtotal: 850.00

Discount: 85.00

Amount after discount: 765.00

Tax: 38.25

Final total cost: 803.25

```
====== RESTART: C:/Users/sivak/Desktop/FODS LAB experiments codes/exp 6.py =====
Subtotal: 850.00
Discount: 85.00
Amount after discount: 765.00
Tax: 38.25
Final total cost: 803.25
```

Result

Thus, the subtotal, discount, and tax were calculated using arithmetic operations, and the customer's final purchase cost was 803.25.